THE ASCENDING CHAIN CONDITION HOLDS FOR QUASI-COHERENT SUBMODULES OF A COHERENT MODULE ON A NOETHERIAN ALGEBRAIC SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- An affine etale cover of a quasi-compact algebraic space. Properties of Spaces, label spaces-properties-lemma-quasi-compact-affine-cover (invoked at proof line 2327). It provides the affine scheme U with a surjective etale morphism $U \to X$, on which the chain can be tested.
- The source of a morphism to a Noetherian space, locally of finite type, is Noetherian. Morphisms of Spaces, label spaces-morphisms-lemma-locally-finite-type-locally-noetherian (invoked at proof line 2330). It makes U a Noetherian scheme, so that the ascending chain condition for coherent modules on U applies.
- The statement for schemes. Cohomology of Schemes, label coherent-lemma-acc-coherent (invoked at proof line 2334). It supplies the ascending chain condition on the Noetherian scheme U, which is then translated back to X.

Source reference. Original lemma 01Y8: sources/stacks-chow/tags.txt, source lines 2547–2547. Navigation pointer for \ref{coherent-lemma-acc-coherent} used inside the counted span (line 2335); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZK: sources/stacks-chow/tags.txt, source lines 6468–6468. Navigation pointer for \ref{spaces-morphisms-lemma-locally-finite-type-locally-noetherian} used inside the counted span (line 2331); the argument that consumes it is retained in the span.

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Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-affine-cover} used inside the counted span (line 2328); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. The ascending chain condition holds for quasi-coherent submodules of $\mathcal{F}$. In other words, given any sequence*

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \ldots \subset \mathcal{F}$$

of quasi-coherent submodules, then $\mathcal{F}_n = \mathcal{F}_{n+1} = \ldots$ for some $n \geq 0$.

Proof. Choose an affine scheme U and a surjective étale morphism $U \to X$ (see Properties of Spaces, Lemma 03H6). Then U is a Noetherian scheme (by Morphisms of Spaces, Lemma 04ZK). If $\mathcal{F}_n|_U = \mathcal{F}_{n+1}|_U = \ldots$ then $\mathcal{F}_n = \mathcal{F}_{n+1} = \ldots$. Hence the result follows from the case of schemes, see Cohomology of Schemes, Lemma 01Y8. □